

Derivatives

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$$

$$\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}$$

Moments and Centroid

$$M_x = \frac{\rho}{2} \int_a^b (f(x)^2 - g(x)^2) dx$$

$$M_y = \rho \int_a^b x(f(x) - g(x)) dx$$

$$m = \rho \int_a^b (f(x) - g(x)) dx$$

$$(\bar{x}, \bar{y}) = \left(\frac{M_y}{m}, \frac{M_x}{m} \right)$$

Integrals

$$\int u dv = uv - \int v du$$

$$\int \tan x dx = \ln |\sec x| + C$$

$$\int \sec x dx = \ln |\sec x + \tan x| + C$$

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

Polar Coordinates

$$\frac{dy}{dx} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}$$

$$A = \int_a^b \frac{1}{2} (f(\theta)^2 - g(\theta)^2) d\theta$$

$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta} \right)^2} d\theta$$

Surface Area

$$A = \int_a^b 2\pi f(x) \sqrt{1 + f'(x)^2} dx$$

$$A = \int_a^b 2\pi x \sqrt{1 + f'(x)^2} dx$$

Area Between Curves

$$A = \int_a^b (f(x) - g(x)) dx$$

Work

$$W = \int_a^b F(x) dx$$

Hydrostatic Force

$$F = \rho g \int_a^b w(y) d(y) dy$$

Arc Length

$$L = \int_a^b \sqrt{1 + f'(x)^2} dx$$

Approximate Integration

$$|E_T| \leq \frac{K(b-a)^3}{12n^2}$$

$$|E_M| \leq \frac{K(b-a)^3}{24n^2}$$

$$|E_S| \leq \frac{L(b-a)^5}{180n^4}$$

Volumes

$$\text{Cross sections: } V = \int_a^b A(x) dx$$

$$\text{Cylindrical shells: } V = \int_a^b 2\pi x f(x) dx$$

Hyperbolic Functions

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$$

$$\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1})$$

$$\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

Trig & Hyperbolic Identities

$$\cos^2 x + \sin^2 x = 1$$

$$\tan^2 x + 1 = \sec^2 x$$

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$1 - \tanh^2 x = \operatorname{sech}^2 x$$

